Trigonometry & Polar Coordinates

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the polar coordinates $\left(6, \frac{5\pi}{4}\right)$ into rectangular Cartesian coordinates (x, y).

Step 1: Apply coordinate conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{5\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right) = -3\sqrt{2}$ Step 2: Compute y: $y = r\sin\theta = 6\sin\left(\frac{5\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right) = -3\sqrt{2}$ Thus the rectangular coordinates are $\boldsymbol{\left(-3\sqrt{2}, -3\sqrt{2}\right)}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the Cartesian equation $x^2 + (y-4)^2 = 16$ into a polar equation relating r and θ.

Step 1: Expand the binomial: $x^2 + y^2 - 8y + 16 = 16 \Rightarrow x^2 + y^2 - 8y = 0$. Step 2: Substitute polar identities $x^2 + y^2 = r^2$ and $y = r\sin\theta$: $r^2 - 8r\sin\theta = 0 \Rightarrow r(r - 8\sin\theta) = 0$. Step 3: Divide by r for $req0$ (the origin is included at $\theta = 0$): $\boldsymbol{r = 8\sin\theta}$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the trigonometric limit: $\lim_{x\to 0} \frac{\sin(5x)}{\tan(2x)}$

Step 1: Express $\tan(2x)$ in terms of sine and cosine: $\frac{\sin(5x)}{\tan(2x)} = \frac{\sin(5x)}{\frac{\sin(2x)}{\cos(2x)}} = \frac{\sin(5x)}{\sin(2x)} \cdot \cos(2x)$ Step 2: Multiply and divide by appropriate factors of x: $\frac{\sin(5x)}{5x} \cdot 5x \cdot \frac{2x}{\sin(2x)} \cdot \frac{1}{2x} \cdot \cos(2x) = \frac{\sin(5x)}{5x} \cdot \frac{1}{\frac{\sin(2x)}{2x}} \cdot \frac{5}{2} \cdot \cos(2x)$ Step 3: Evaluate limits using $\lim_{u\to 0} \frac{\sin u}{u} = 1$ and $\cos(0) = 1$: $(1) \cdot (1) \cdot \frac{5}{2} \cdot 1 = \boldsymbol{\frac{5}{2}}$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation on the interval $[0, 2\pi)$: $2\cos^2 x - 3\cos x + 1 = 0$

Step 1: Factor as a quadratic in $\cos x$: $(2\cos x - 1)(\cos x - 1) = 0$ Step 2: Solve each factor: 1. $2\cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow \boldsymbol{x = \frac{\pi}{3}, \quad \frac{5\pi}{3}}$. 2. $\cos x - 1 = 0 \Rightarrow \cos x = 1 \Rightarrow \boldsymbol{x = 0}$. All solutions in $[0, 2\pi)$: $\boldsymbol{0, \quad \frac{\pi}{3}, \quad \frac{5\pi}{3}}$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all complex solutions to the cubic equation $z^3 + 8 = 0$.

Step 1: Rewrite as a difference of cubes: $z^3 - (-8) = 0 \Rightarrow z^3 + 2^3 = (z+2)(z^2 - 2z + 4) = 0$. Step 2: Real root: $z + 2 = 0 \Rightarrow \boldsymbol{z_1 = -2}$. Step 3: Complex roots via quadratic formula: $z^2 - 2z + 4 = 0 \Rightarrow z = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(4)}}{2} = \frac{2 \pm \sqrt{4 - 16}}{2} = \frac{2 \pm \sqrt{-12}}{2} = \frac{2 \pm 2i\sqrt{3}}{2} = \boldsymbol{1 \pm i\sqrt{3}}$. Thus the three roots are $\boldsymbol{-2, \quad 1 + i\sqrt{3}, \quad 1 - i\sqrt{3}}$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the limit: $\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x}$

Step 1: Rationalize numerator using the conjugate $\sqrt{1+x}+1$: $\lim_{x\to 0}\frac{(\sqrt{1+x}-1)(\sqrt{1+x}+1)}{x(\sqrt{1+x}+1)}=\lim_{x\to 0}\frac{(1+x)-1}{x(\sqrt{1+x}+1)}$ Step 2: Cancel x and evaluate: $\lim_{x\to 0}\frac{x}{x(\sqrt{1+x}+1)}=\lim_{x\to 0}\frac{1}{\sqrt{1+x}+1}=\frac{1}{\sqrt{1+0}+1}=\mathbf{\frac{1}{2}}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the scalar projection of $\mathbf{u}=\langle 4,3\rangle$ onto $\mathbf{v}=\langle 12,5\rangle$.

Formula: $\text{comp}_{\mathbf{v}}\mathbf{u}=\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|}$. Step 1: Compute $\mathbf{u}\cdot\mathbf{v}$: $\mathbf{u}\cdot\mathbf{v}=(4)(12)+(3)(5)=48+15=63$. Step 2: Compute $|\mathbf{v}|$: $|\mathbf{v}|=\sqrt{12^2+5^2}=\sqrt{144+25}=\sqrt{169}=13$. Step 3: Calculate scalar projection: $\text{comp}_{\mathbf{v}}\mathbf{u}=\mathbf{\frac{63}{13}}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Simplify the complex quotient into standard form $a+bi$: $\frac{3+4i}{1+2i}$

Step 1: Multiply numerator and denominator by the conjugate $1-2i$: $\frac{3+4i}{1+2i}=\frac{(3+4i)(1-2i)}{(1+2i)(1-2i)}$ Step 2: Expand numerator and denominator: Numerator: $3(1)-4(-2)+[3(-2)+4(1)]i=3+8+(-6+4)i=11-2i$. Denominator: $1^2+2^2=1+4=5$. Step 3: Separate real and imaginary parts: $\frac{11-2i}{5}=\mathbf{\frac{11}{5}-\frac{2}{5}i}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the power of the imaginary unit: $i^{2026}+i^{2027}+i^{2028}+i^{2029}$

Step 1: Recall the periodicity of powers of i: Powers of i repeat cyclically with period 4: $i^1=i$, $i^2=-1$, $i^3=-i$, $i^4=1$. The sum of any 4 consecutive integer powers of i is identically zero: $i^1+i^2+i^3+i^4=i-1-i+1=0$. Step 2: Apply to the consecutive sequence: $i^{2026}+i^{2027}+i^{2028}+i^{2029}=(-1)+(-i)+(1)+(i)=\mathbf{0}$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A constant force vector $\mathbf{F}=\langle 8,-3\rangle$ (in newtons) moves an object along the displacement vector $\mathbf{d}=\langle 5,2\rangle$ (in meters). Calculate the work done by the force.

Definition of Work: $W=\mathbf{F}\cdot\mathbf{d}$. Step 1: Evaluate the dot product: $W=(8)(5)+(-3)(2)=40-6=\mathbf{34}$ J. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given that $\sin\theta = \frac{3}{5}$ with θ in Quadrant II, find the exact value of $\sin(2\theta)$.

Step 1: Find $\cos\theta$: In Quadrant II, cosine is negative: $\cos\theta = -\sqrt{1-\sin^2\theta} = -\sqrt{1-\left(\frac{3}{5}\right)^2} = -\sqrt{\frac{16}{25}} = -\frac{4}{5}$. Step 2: Apply double-angle formula: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(-\frac{4}{5}\right) = \mathbf{-\frac{24}{25}}$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the indeterminate limit: $\lim_{x\to 5} \frac{x^2-25}{x^2-8x+15}$

Step 1: Direct substitution check: $\frac{5^2-25}{5^2-40+15} = \frac{0}{0}$ (indeterminate). Step 2: Factor numerator and denominator: $\frac{(x-5)(x+5)}{(x-5)(x-3)}$ Step 3: Cancel common factor and evaluate: $\lim_{x\to 5} \frac{x+5}{x-3} = \frac{5+5}{5-3} = \frac{10}{2} = \mathbf{5}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, $A = 45^\circ$, $B = 60^\circ$, and $a = 8$. Find the exact length of side b.

Step 1: Apply Law of Sines: $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{a\sin B}{\sin A}$ Step 2: Substitute exact values: $b = \frac{8\sin 60^\circ}{\sin 45^\circ} = \frac{8\left(\frac{\sqrt{3}}{2}\right)}{\frac{\sqrt{2}}{2}} = \frac{4\sqrt{3}}{\frac{\sqrt{2}}{2}} = \frac{8\sqrt{3}}{\sqrt{2}} = \mathbf{4\sqrt{6}}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve for x: $\log_2(x) + \log_2(x-6) = 4$

Step 1: Domain restrictions: $x > 0$ and $x - 6 > 0 \Rightarrow x > 6$. Step 2: Apply product rule for logs: $\log_2(x(x-6)) = 4 \Rightarrow x(x-6) = 2^4 = 16$ $x^2 - 6x - 16 = 0 \Rightarrow (x-8)(x+2) = 0$. Step 3: Check domain: $x = -2$ is extraneous (rejected). Therefore $\mathbf{x = 8}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Vectors $\mathbf{u} = \langle 3, -2, 5\rangle$ and $\mathbf{v} = \langle 4, k, -2\rangle$ are orthogonal in $\mathbb{R}^3$. Find the value of k.

Step 1: Orthogonality criterion: Two nonzero vectors are orthogonal if and only if their dot product equals 0: $\mathbf{u}\cdot\mathbf{v} = 0$ Step 2: Compute dot product: $(3)(4) + (-2)(k) + (5)(-2) = 0$ $12 - 2k - 10 = 0 \Rightarrow 2 - 2k = 0 \Rightarrow 2k = 2 \Rightarrow \mathbf{k = 1}$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

Evaluate the limit at infinity: $\lim_{x\to\infty} \frac{5x^3-2x+7}{2x^3+4x^2-1}$

Step 1: Compare degrees of numerator and denominator: Both numerator and denominator have degree 3. Step 2: Apply Theorem 6.1 (Ratio of leading coefficients): $\lim_{x\to\infty} \frac{5x^3-2x+7}{2x^3+4x^2-1} = \frac{5}{2} = \mathbf{\frac{5}{2}}$.

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Evaluate the limit: $\lim_{x \to 2} \frac{3x^2 - 4x + 1}{x + 3}$

Since the rational function is continuous at $x = 2$ (denominator $2 + 3 = 5 eq 0$), evaluate by direct substitution: $\lim_{x \to 2} \frac{3x^2 - 4x + 1}{x + 3} = \frac{3(2^2) - 4(2) + 1}{2 + 3} = \frac{12 - 8 + 1}{5} = \frac{5}{5} = \mathbf{1}$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Solve the exponential equation for x: $4^{x+1} = 8^{2x-1}$

Step 1: Express both sides using base 2: $4 = 2^2 \Rightarrow 4^{x+1} = (2^2)^{x+1} = 2^{2x+2}$

$8 = 2^3 \Rightarrow 8^{2x-1} = (2^3)^{2x-1} = 2^{6x-3}$ Step 2: Equate exponents: $2x + 2 = 6x - 3 \Rightarrow 5 = 4x \Rightarrow \mathbf{x = \frac{5}{4}}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

Given $A = (2, -1; 3, 4)$ and $B = (0, 5; -2, 1)$, compute $3A - 2B$.

Step 1: Compute scalar multiples: $3A = (6, -3; 9, 12)$, $2B = (0, 10; -4, 2)$ Step 2: Subtract entrywise: $3A - 2B = (6 - 0, -3 - 10; 9 - (-4), 12 - 2) = (\mathbf{6, -13; 13, 10})$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Use the Remainder Theorem to find the remainder when $P(x) = 3x^4 - 2x^3 + 5x - 7$ is divided by $x + 2$.

Remainder Theorem: The remainder upon division by $x - c$ is $P(c)$. Here $c = -2$:

$P(-2) = 3(-2)^4 - 2(-2)^3 + 5(-2) - 7$

$= 3(16) - 2(-8) - 10 - 7 = 48 + 16 - 10 - 7 = 64 - 17 = \mathbf{47}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $2\sin(5x)\cos(3x)$ as a sum of trigonometric functions.

a. $\sin(8x) + \sin(2x)$ ✓

b. $\sin(8x) - \sin(2x)$

c. $\cos(8x) + \cos(2x)$

d. $\cos(8x) - \cos(2x)$

Using the product-to-sum identity $2\sin A\cos B = \sin(A + B) + \sin(A - B)$ gives $\sin(8x) + \sin(2x)$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the expression $\frac{\sin(3x)+\sin(x)}{\cos(3x)+\cos(x)}$.

a. $\tan(2x)$ ✓

b. $\tan(x)$

c. $\cot(2x)$

d. $\tan(4x)$

Using sum-to-product: numerator $= 2\sin(2x)\cos(x)$ and denominator $= 2\cos(2x)\cos(x)$. Dividing gives $\frac{\sin(2x)}{\cos(2x)} = \tan(2x)$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the exact value of $\cos(105^\circ) - \cos(15^\circ)$?

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{6}}{2}$

d. $\frac{\sqrt{6}}{2}$

Sum-to-product: $-2\sin(60^\circ)\sin(45^\circ) = -2(\frac{\sqrt{3}}{2})(\frac{\sqrt{2}}{2}) = -\frac{\sqrt{6}}{2}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What geometric curve is represented by the polar equation $r = 2 + 2\cos\theta$?

a. **Cardioid** ✓

b. Limaçon with inner loop

c. Three-petal rose

d. Circle

An equation of the form $r = a + a\cos\theta$ has equal coefficients ($a = b$), which represents a cardioid (heart-shaped curve).

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 ✓

b. 6

c. 4

d. 8

For a rose curve $r = a\cos(n\theta)$, if n is odd, the curve has n petals. Here $n = 3$, so it has 3 petals.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 4

b. 8 ✓

c. 16

d. 2

If n is even in $r = a\sin(n\theta)$, the curve has $2n$ petals. Here $n = 4$, so it has $2(4) = 8$ petals.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar point $(4, \frac{5\pi}{6})$ to rectangular coordinates (x, y).

a. $(-2\sqrt{3}, 2)$ ✓

b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(2, -2\sqrt{3})$

$x = r\cos\theta = 4\cos(\frac{5\pi}{6}) = 4(-\frac{\sqrt{3}}{2}) = -2\sqrt{3}$. $y = r\sin\theta = 4\sin(\frac{5\pi}{6}) = 4(\frac{1}{2}) = 2$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular point $(-3, -3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(3\sqrt{2}, \frac{5\pi}{4})$ ✓

b. $(3\sqrt{2}, \frac{3\pi}{4})$

c. $(6, \frac{5\pi}{4})$

d. $(3\sqrt{2}, \frac{7\pi}{4})$

$r = \sqrt{(-3)^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$. Point is in Quadrant III: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the Cartesian equation for the polar line $\theta = \frac{\pi}{3}$?

a. $y = \sqrt{3}x$ ✓

b. $y = \frac{\sqrt{3}}{3}x$

c. $y = x + \frac{\pi}{3}$

d. $x = \sqrt{3}y$

Take tangent of both sides: $\tan\theta = \tan(\frac{\pi}{3}) = \sqrt{3}$. Since $\tan\theta = \frac{y}{x}$, we have $\frac{y}{x} = \sqrt{3} \Rightarrow y = \sqrt{3}x$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $\cos(3x) = \frac{1}{2}$ for $x \in [0, \pi]$.

a. $\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{9}, \frac{7\pi}{9}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

For $x \in [0, \pi]$, $3x \in [0, 3\pi]$. $\cos(3x) = 1/2 \Rightarrow 3x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$. Dividing by 3 gives $x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all solutions to $\sin(2x) = \cos(x)$ in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

b. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{6}, \frac{5\pi}{6}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$. $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the range of $f(x) = 2 - 3\sin(4x)$?

a. $[-1, 5]$ ✓

b. $[-3, 3]$

c. $[-5, 1]$

d. $[2, 5]$

Since $-1 \le \sin(4x) \le 1$, we have $-3 \le -3\sin(4x) \le 3$. Adding 2 gives $-1 \le f(x) \le 5$.

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33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y)-coordinate plane, what are the rectangular coordinates of the point with polar coordinates $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$?

a. $(-3, 3\sqrt{3})$

b. $(-3\sqrt{3}, 3)$ ✓

c. $(3\sqrt{3}, -3)$

d. $(-3\sqrt{3}, -3)$

To convert from polar coordinates (r, θ) to rectangular coordinates (x, y), use the conversion formulas: $x = r\cos\theta$ and $y = r\sin\theta$ Given $r = 6$ and $\theta = \frac{5\pi}{6}$: $x = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$ $y = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$ Thus, the rectangular coordinates are $(-3\sqrt{3}, 3)$. - Choice A $(-3, 3\sqrt{3})$ results from inadvertently reversing the sine and cosine components (calculating $x = r\sin\theta$ and $y = r\cos\theta$). - Choice C $(3\sqrt{3}, -3)$ results from applying Quadrant IV signs instead of Quadrant II signs. - Choice D $(-3\sqrt{3}, -3)$ results from incorrectly treating both coordinates as negative (Quadrant III).

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A point in the polar coordinate plane has coordinates $(r, \theta) = \left(4, \frac{5\pi}{6}\right)$. Which of the following gives the rectangular coordinates (x, y) of this point?

a. $(-2\sqrt{3}, 2)$ ✓

b. $(-2, 2\sqrt{3})$

c. $(2, -2\sqrt{3})$

d. $(2\sqrt{3}, -2)$

To convert polar coordinates (r, θ) to rectangular coordinates (x, y), use the conversion formulas: $x = r\cos\theta$ and $y = r\sin\theta$ Given $r = 4$ and $\theta = \frac{5\pi}{6}$: $x = 4\cos\left(\frac{5\pi}{6}\right) = 4\left(-\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$ $y = 4\sin\left(\frac{5\pi}{6}\right) = 4\left(\frac{1}{2}\right) = 2$ Thus, the rectangular coordinates are $(-2\sqrt{3}, 2)$. - Choice B is incorrect because it mistakenly swaps the values of $\cos\left(\frac{5\pi}{6}\right)$ and $\sin\left(\frac{5\pi}{6}\right)$. - Choice C is incorrect because it swaps the x- and y-coordinates. - Choice D is incorrect because it reverses the signs of both coordinates.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3.464, -2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4, 210^\circ)$ ✓

b. $(4, 30^\circ)$

c. $(4, 150^\circ)$

d. $(8, 210^\circ)$

$r = \sqrt{x^2 + y^2} \approx 4$; the reference angle and the quadrant of $(-3.464, -2)$ give $\theta = 210^\circ$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3,\ -5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6,\ 240^\circ)$ ✓

b. $(6,\ 60^\circ)$

c. $(6,\ 120^\circ)$

d. $(12,\ 240^\circ)$

$r = \sqrt{x^2 + y^2} \approx 6$; the reference angle and the quadrant of $(-3,\ -5.196)$ give $\theta = 240^\circ$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3.464,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 150^\circ)$ ✓

b. $(4,\ 330^\circ)$

c. $(4,\ 210^\circ)$

d. $(8,\ 150^\circ)$

$r = \sqrt{x^2 + y^2} \approx 4$; the reference angle and the quadrant of $(-3.464, 2)$ give $\theta = 150^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Convert the rectangular point $(3.464,\ -2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 330^\circ)$ ✓

b. $(4,\ 150^\circ)$

c. $(4,\ 30^\circ)$

d. $(8,\ 330^\circ)$

$r = \sqrt{x^2 + y^2} \approx 4$; the reference angle and the quadrant of $(3.464,\ -2)$ give $\theta = 330^\circ$.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

Convert the rectangular point $(-3,\ 5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6,\ 120^\circ)$ ✓

b. $(6,\ 300^\circ)$

c. $(6,\ 240^\circ)$

d. $(12,\ 120^\circ)$

$r = \sqrt{x^2 + y^2} \approx 6$; the reference angle and the quadrant of $(-3, 5.196)$ give $\theta = 120^\circ$.

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Functions & Equations

Convert the rectangular point $(3,\ -5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6,\ 300^\circ)$ ✓

b. $(6,\ 120^\circ)$

c. $(6,\ 60^\circ)$

d. $(12,\ 300^\circ)$

$r = \sqrt{x^2 + y^2} \approx 6$; the reference angle and the quadrant of $(3,\ -5.196)$ give $\theta = 300^\circ$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Limits

Convert the rectangular point $(-2,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(2.83,\ 135^\circ)$ ✓

b. $(2.83,\ 315^\circ)$

c. $(2.83,\ 225^\circ)$

d. $(6,\ 135^\circ)$

$r = \sqrt{x^2 + y^2} \approx 2.83$; the reference angle and the quadrant of $(-2, 2)$ give $\theta = 135^\circ$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Complex Numbers

Convert the rectangular point $(3,\ 5.196)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(6,\ 60^\circ)$ ✓

b. $(6,\ 240^\circ)$

c. $(6,\ 300^\circ)$

d. $(12,\ 60^\circ)$

$r = \sqrt{x^2 + y^2} \approx 6$; the reference angle and the quadrant of $(3, 5.196)$ give $\theta = 60^\circ$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Convert the rectangular point $(3.464,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 30^\circ)$ ✓

b. $(4,\ 210^\circ)$

c. $(4,\ 330^\circ)$

d. $(8,\ 30^\circ)$

$r = \sqrt{x^2 + y^2} \approx 4$; the reference angle and the quadrant of $(3.464, 2)$ give $\theta = 30^\circ$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

Convert the rectangular point $(2,\ 2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(2.83,\ 45^\circ)$ ✓

b. $(2.83,\ 225^\circ)$

c. $(2.83,\ 315^\circ)$

d. $(6,\ 45^\circ)$

$r = \sqrt{x^2 + y^2} \approx 2.83$; the reference angle and the quadrant of $(2, 2)$ give $\theta = 45^\circ$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Functions & Equations

Convert the rectangular point $(-4,\ 0)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4,\ 180^\circ)$ ✓

b. $(4,\ 0^\circ)$

c. $(4,\ 210^\circ)$

d. $(8,\ 180^\circ)$

$r = \sqrt{x^2 + y^2} \approx 4$; the reference angle and the quadrant of $(-4, 0)$ give $\theta = 180^\circ$.

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